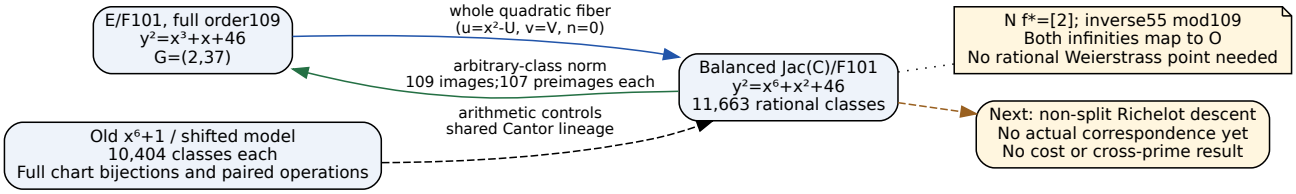
Balanced genus2 arithmetic without a Weierstrass chart

2026-11-11 · Chained ECDLP Transfers · bounded development prototype

Unsigned/unverified standing. The monic split-sextic prototype supports checked primes5..101. Production arithmetic and the1,940-row cover catalog remain unchanged. No benchmark or computational advantage. [pcc:unverified]



Primary representation and reduction

Store a degree2 effective divisor relative to K=∞++∞−: finite(u,v) + n∞+ + (2−deg(u)−n)∞− − K, with monic u, degree≤2, deg(v)<deg(u), u dividing f−v², and 0≤n≤2−deg(u). Identity is(1,0,1); the infinity difference δ is(1,0,2). The inverse sends v to−v mod u and n to2−deg(u)−n. Galbraith corrected Definition10.4.9/Theorem10.4.19 supply uniqueness. [pcc:unverified]

Affine Cantor composition retains the signed infinity count. Reduction uses a cubic ℓ matching the required infinity branch, exact quotient$(f-\ell^2)/u$, and the actual valuation of $y-\ell$. Degree4 interpolation with leading±1 is handled explicitly; treating both poles as order3 there would be incorrect. All output degree/infinity bounds and divisions are checked. [pcc:unverified]

Six core Lean lemmas audit degree/count bookkeeping, inverse counts, branch updates and the infinity-polynomial square. They do not construct valuations or Picard groups. The6-declaration gate passes; crypto PCC-1cadd0 remains unsigned. [pcc:unverified]

Independent arithmetic controls

All10,404 classes of $y^2=x^6+1$/F101 biject with the existing odd chart at(10,0). Paired additions, every inverse/cancellation, δ-translation and first quotient norm agree. All six degree/infinity patterns occur; δ has order3. All10,404 classes of $y^2=(x+1)^6+1$ also match after translation, with nonzero infinity polynomial x^3+3x^2+3x+1. Both engines use Cantor; this is producer-run agreement, not signed scientific validation. [pcc:unverified]

Full prime-order transfer and unresolved scope

The existing E109/F101 source $y^2=x^3+x+46$ pulls back through$(x,y)\mapsto(x^2,y)$ to $C:y^2=x^6+x^2+46$ without rational Weierstrass points. Every source scalar passes pullback/norm[2] and inverse55 recovery. Point recovery does not compute the unknown DLP scalar. [pcc:unverified]

The balanced Jacobian has11,663 classes. Separate elliptic point counts give109 and107 for the source and normalized reciprocal quotient $y^2=x^3+x^2+96$; the product agrees. All11,663 paired arbitrary-class norm additions pass. The norm has109 images and equal fibers107. These are rational counts, not geometric kernel degrees. [pcc:unverified]

The primary F37 example $y^2=x^6+x+1$ reproduces Galbraith Example10.4.18; 1,211 fixed associativity triples and malformed/foreign-model controls pass. Exact source: src/balanced.rs. The closed-point norm helper is shared with the earlier cubic prototype, whose tests pass again after the refactor. [pcc:unverified]

Receipts and reproduction

Run experiments/chained-transfer-reference/check.sh with Rust1.99.1 on PATH and the immutable upstream75a737b8 runtime closure available. Receipt balanced-pinned-01.txt binds source guards, formatting, strict Clippy, native25 and reference37 passing tests, with1 pre-existing benchmark ignored. PCC-8b7ca0 remains unsigned. First compile and Clippy failures are retained; no prior receipt, witness or test was rewritten. [pcc:unverified]

Full written argument: BALANCED.md. Primary source: Galbraith corrected ch.10§10.4.2, especially Definition10.4.9, Lemmas10.4.10/14, Theorem10.4.19 and Example10.4.18; corrected ch.8§8.3 for Picard norm. Crypto’s SOURCES_20261111.md binds the primary PDF bytes and retrieval gaps.

What remains open

This bounded prime-field prototype does not implement large-prime or extension-field arithmetic, nonmonic/inert models, catalog replay, normalized Montgomery/Edwards targets or production integration. It narrows the catalog problem without closing it. [pcc:unverified]

Next: prove and construct Frobenius-stable nonrational Richelot splittings on predeclared E19/F17, then implement the actual correspondence and descent. No such correspondence is supplied here. The old rational-factor constructor restriction is not a universal kernel obstruction. [pcc:unverified]

S3 admission, live IsoLab budget/capacity, KB and the encrypted atlas remain unavailable. No solver comparison, relation collector or paid job ran. Crypto PR3877 remains blocked by frozen-ledger/history and archived-receipt requirements. Canonical transfer/leaderboard graphs are unchanged because no signed finding or performance edge is promoted. Source queue: crypto3904.